

LESSON 9-1 • SESSION 1 • PRACTICE • FORM A

Independent and Dependent Variables

Learning target – I can name the independent and dependent variables in a relationship. · 6.EE.C.9

REMEMBER

- The **independent variable** is the input: you choose it, or it comes first.
- The **dependent variable** is the output. It changes because of the input.
- Sentence test: “**dependent** depends on **independent**.” Independent goes on the x-axis.

WORKED EXAMPLE

A bike rental costs \$5 for each hour. Name the variables.

- Step 1** The quantities are hours rented and total cost.
Step 2 Cost changes when hours change.
Step 3 “Total cost depends on hours rented” makes sense.

Answer hours = independent; cost = dependent

WORDS independent variable (*variable independiente*) · dependent variable (*variable dependiente*) · depends on (*depende de*)

PRACTICE

show your work · label units

1. For each pair, write **I** under the independent variable and **D** under the dependent variable.

	Quantity	I or D	Quantity	I or D
a.	hours worked		money earned	
b.	test score		hours studied	
c.	miles driven		gallons of fuel	
d.	guests at a party		pizzas needed	

2. **Popcorn Stand.** Nia sells popcorn at games. The more bags she sells, the more money she collects. Longer games lead to more bags sold.
- a. Bags sold and money collected. Independent: _____ Dependent: _____
- b. Game length and bags sold. Independent: _____ Dependent: _____
- c. Which quantity is independent in one pair and dependent in the other? _____
3. **Science Fair Ferns.** Theo waters some ferns more often than others. Then he measures how tall each fern grows.
- Hint** What does the student control? What is only measured?
- a. Independent variable: _____
- b. Dependent variable: _____
4. **Bus Ride.** The farther a bus travels, the more riders board. A student says the number of riders is the independent variable. Do you agree?
- Hint** Use the sentence test: what depends on what?
- Agree?** _____
5. **Dog Walking.** Jaden walks dogs. The more hours he walks, the more money he earns. Name the dependent variable.
- Hint** Which one changes because of the other?
- Dependent** _____

6. **Which Axis?** A graph shows the cost of a taxi ride and the number of miles driven.

Hint The independent variable goes across.

a. Which variable goes on the x-axis? _____

b. Which variable goes on the y-axis? _____

7. **Downloading Photos.** The more photos Priya picks, the longer the download takes. Write a sentence with "depends on." Then name the dependent variable.

Hint Priya picks the photos; the time only responds.

Sentence _____

8. **Sneaker Sale.** Every pair of sneakers costs the same price. A student says total money collected is the independent variable because it is the bigger number. Explain whether you agree. Then name both variables.

Show your work.

Variables _____

FIND THE ERROR

explain the mistake, then fix it

9. **Marisol's Work.** A movie theater sells drinks. Marisol wrote: "The number of drinks sold depends on the money the theater collects." Explain her mistake and write a correct sentence.

TEST PRACTICE

choose the best answer

10. The independent variable goes on the –

A x-axis

B y-axis

C origin

D title

11. In "hours worked and money earned," the dependent variable is –

A hours worked

B money earned

C both

D neither

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

Disagree	bags sold	height	time	miles	Agree	money	ferns	dogs	hours
T	O	U	T	U	Q	P	Z	J	X
		#2c	#3b	#4	#5	#6a	#7		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____